

synaptic — Technical Reference

synaptic project

CTAN Community

Abstract: Technical overview of synaptic v2: module architecture, namespace conventions, mode dispatch, and extension points.

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1 Architecture

1.1 Module Dependency Graph

synaptic.sty (thin shell, RequirePackageWithOptions -> base)

- synaptic-base.sty Engine check, option parsing
- synaptic-color.sty Theme colour system (5 themes)
- synaptic-fonts.sty Fontset selection + CJK support
 - | --- synaptic-fonts.lua Lua font auto-detection
- synaptic-layout.sty Page geometry, microtype, headings
- synaptic-title.sty Title page rendering (journal)
- synaptic-theorem.sty Theorem/proof environments
- synaptic-book.sty Book mode (chapter, headers)
- synaptic-lecture.sty Lecture mode (wide margins)
- synaptic-boxes.sty tcolorbox environments
- synaptic-lang-en.def English labels
- synaptic-lang-zh.def Chinese labels

1.2 Mode Dispatch Table

mode	Document Class	Loaded Modules
journal	scrartcl	base, color, fonts, layout, title, theorem
book	scrbook	journal modules + book
lecture	scrartcl / scrreprt	base, color, fonts, layout, theorem, lecture, boxes
notes	scrartcl	base, color, fonts, layout, theorem

1.3 Namespace Convention

Scope	Pattern	Example
User API	\Synaptic*	\SynapticTitle
User environments	synaptic*	synapticexample
Internal macros	__synaptic_*	__synaptic_title_render:
Local variables	\l_synaptic_*	\l_synaptic_mode_tl
Label constants	\c_synaptic_label_*	\c_synaptic_label_thm_tl

2 Key Commands Reference

2.1 User-Facing API

`\SynapticSetup{key=val}` Runtime option changes.

`\SynapticTitle{text}` Document title.

`\SynapticAuthor[mark]{name}` Author with optional affiliation mark.

`\SynapticAffiliation[mark]{text}` Affiliation.

`\SynapticEmail{addr}` Email address.

`\SynapticKeywords{...}` Keywords below abstract.

`\SynapticArxiv{ID}` arXiv identifier link.

`\SynapticHeader{text}` Custom submission header.

`\SynapticSubHeader{text}` Sub-header.

`\SynapticPreprint{ID}` Preprint identifier.

`\SynapticDedicated{text}` Dedication text.

`\SynapticMakeTitle` Render title page.

`\SynapticNewTheorem{env}{Name}` Declare theorem-like environment.

`\SynapticDefineTheme{primary}{secondary}` Custom colour theme.

2.2 Environments

abstract Collect abstract text for title page.

theorem, lemma, proposition, definition Theorem environments (shared counter, numbered by section).

proof Proof environment with QED symbol.

example, remark, warning, exercise tcolorbox environments (lecture mode).

synbox Generic tcolorbox with optional title.

3 Extension Points

- **New languages** — Copy `synaptic-lang-en.def` to `synaptic-lang-fr.def` and translate each label constant.
- **New themes** — Use `\SynapticDefineTheme{name}{primary}{secondary}`.
- **New font sets** — Add a new case to `synaptic-fonts.sty` in the dispatch block.
- **New modes** — Create a new `synaptic-*.sty` file and add a case to the mode dispatch in `synaptic-base.sty`.
- **Font auto-detect** — Edit `synaptic-fonts.lua` to modify the detection priority list.

4 Version History

v2.0 (2026-06-05) Complete modular refactoring: mode system, theme system, fontset abstraction, Lua font detection, non-invasive API, tcolorbox environments.

v1.0 (2025) Initial monolithic release: single `synaptic.sty` with basic colour/style options.